Fiscal Year:	FY 2006	**Task Last Updated:**	FY 05/18/2006
PI Name:	Steen, Paul H Ph.D.		
Project Title:	Dynamics and Stability of Capillary Systems: Moving Liquids by Design		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Dynamics and stability		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	PI is deceased as of September 2020		
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Fluid Physics 01-OBPR-02: Research and Flight Experiment Opportunities
Start Date:	02/11/2002	**End Date:**	11/30/2005
No. of Post Docs:	2	**No. of PhD Degrees:**	1
No. of PhD Candidates:	0	**No. of Master' Degrees:**	1
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No. of Bachelor's Candidates:	7	**Monitoring Center:**	NASA GRC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
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Key Personnel Changes/Previous PI:	Steen, Paul H.		
COI Name (Institution):	Hirsa, Amir (Rensselaer Polytechnic Institute)		
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	Human Exploration and Development of Space (HEDS) requires efficient, reliable and lightweight technologies. The dynamics and stability of interfaces and systems of interfaces (two-phase and three-phase multi-component systems) is central to a range of these technologies. The maximum length-scale below which surface tension controls the shape of gas/liquid interfaces increases from 1 mm on earth to about 1 m in orbit. The premise is that **dramatic shape changes can be manipulated to advantage with minimal input, if the system is near instability**. The primary objective is to develop the science base to allow novel approaches to liquid management in low-gravity. To focus the effort, both short-term and long-term challenges are set forth. On the short term, the technological goal is a **substrate surface with controllable nearly reversible wetting characteristics**. On the long-term, the technological goal is to design a **small-scale liquid transport device (a novel pump) with low energy (and voltage) input and whose only moving parts		

Task Description:	involve the motion of liquids**. This would represent an isothermal system with multiple stable states, amongst which nearly-reversible state changes can occur and whose effect can be amplified by the action of elements in parallel. Objectives of the theory and computations are 1) to provide the equilibrium response diagrams (p-v) for capillary elements, combinations of elements (including switches), and arrays of elements (including arrays of switches), 2) to provide the kinetic response of various switches (and arrays of switches) of choice, 3) to predict the robustness of various switches (basins of attraction) and 4) to answer "to what extent are dynamical trajectories influenced by equilibrium energy landscapes" for capillary systems. This will provide support for and guide the experimental work. Experimental objectives are to construct and test capillary elements, switches, arrays of switches and capillary devices for their equilibrium response, energy barriers and kinetics of switching. Testing of millimeter-scale (liquid/gas) and centimeter scale (liquid/liquid with density balance) will largely be the task of RPI. Testing on the centimeter scale with a soap-film analogy will be the responsibility of Cornell. The dynamics and stability of interfaces and systems of interfaces is central to a range of HEDS technologies. Novel approaches to liquid management in low gravity will support the technological challenges of long-term missions, manned or unmanned. A successful outcome of the proposed work will yield a substrate with controllable nearly reversible wetting characteristics (see 'controlled adhesion' proof-of-concept VC) which is a key step in controlling the position of liquids in two-phase liquid/gas configurations in low-gravity. Furthermore, a foundation will be provided for a novel pump that moves liquids by droplets (liquid packets held by surface tension), much like in a "bucket brigade" (see 'droplet mover' proof-of-concept VC). The idea is that gas-lubricated droplet motions will involve much less dissipation than liquid motions adjacent to a solid surface. Small-capacity and low-power devices such as sensors and monitors are the HEDS applications that are most likely to benefit.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Although the premium on devices of low-weight requiring low-power make our goals especially beneficial to long duration space missions, the benefits will also accrue earth-based applications. The capillary length scale decreases to about a millimeter on earth. This means that capillary elements that make up controllable substrates with nearly reversible wetting characteristics will need to be sub-millimeter size. Otherwise, all paradigms for moving liquids and controlling adhesion and wetting for low gravity environments will be applicable also on earth.
	The droplet switch: energy landscapes and equilibrium behavior. The droplet switch has a double-well energy landscape. A millimeter-scale prototype using water is studied. Its response diagram and energy barriers are probed using two different activation (or triggering) methods [3]. A pressure difference is applied across the switch in the pressure-activation method while the surface tension of one droplet is modified by a redox-active surfactant in the electrochemical-activation method. The first corresponds to kicking the system from one well to another while the second corresponds to modifying the wells relative to one-another until one destabilizes. Although switching is observed by both means, pressure-activation results in much faster toggling [P4]. The observations from pressure-activation show good agreement with the theoretically predicted equilibrium states (bifurcation diagram) and energy barrier amplitude [3]. Liquid/solid wall contacting: towards an adhesion device. Using pressure-activated switches in the presence of an obstructing wall provides a second means of testing equilibrium behavior. Here, the wall is placed orthogonal to the line of action of the switch and attachment by the droplet to the wall may or may not occur depending on separation distance (see 'wall-connection' video clip, URL at top). The observed behavior is consistent with theory [P17]. Robustness of switches is related to the fate of finite-amplitude disturbances and the basins-of-attraction for the competing energy wells. The establishment of 'trapping regions' gives the fate of certain finite-amplitude disturbances[8]. This method has been introduced for a related problem, also showing double energy wells [1]. It indicates to what extent the dynamical trajectories are influenced by equilibrium energy landscapes. Switches with up to quintuplet-wells have been predicted [5]. Electroosmotic actuation: concept. Pressure-activation is fast but awkward for addressing switches at the millimeter scale and below. Electrochemical-activation can address switches individually but is slow and lacks robustness (to date) [3]. An alternative activation means was sought. An electrokinetically-activated droplet switch was conceived, built and reduced to practice (patent application). The principle uses an electoosmotic pump to move liquid from one droplet to the other [4, P8-10, P12]. It is placed in-between the two droplets (schematic below). The resulting switch (or actuator, more broadly) has no moving solid parts, uses low voltage, has switching times less than one second and scales favorably to droplet sizes approaching 1 micron [4]. See 'electro-osmotic droplet switch (EODS)' video clip (URL at top). Actuator realization [4]. With millimeter-size water droplets and micron-size pores in a glass frit, 5 volts gives time-to-switch under 1 second – before optimization. The figure shows the device used. Switching frequency of 50 Hz has been demonstrated. See "Electro-osmostic see-saw action" video clip (URL at top). Switching time scales favorablypromising faster response from lower voltage [4]. For 2B = 170 micron, for example, a switching time of 25 ms is predicted for 5 V using a typical pore size of 2R = 1.2 micron. Thus, frequencies better than 100 Hz are within reach. Although the zeta potential of water against glass is key to the switching strength, other liquids against polymers will work. Furthermore, since liquid is only necessary within the pores, a second immiscible liquid (e.g. low vapor pressure) can be used for the exposed top and bottom droplets. Alternatively, 'elastic switches' are possible, where much of the liquid is replaced by gel, hydrogel or elastic polymer and elasticity plays the role of surface tension. Kinetics. Switching with topological change is important for both technological goals. For example, how long does the breakup take and how much residual droplet is left on a substrate? That is, what is the kinetics of breakup and how does the bridge breakup? This was studied for a capillary sphere stretched to a peanut-shaped initial condition [2, P7]. Time-to-pinchoff and structure at pinchoff are predicted computationally via the vortex-sheet method of simulation of a capillary interface. Publications. 1. Bhandar, A.S., M.J. Vogel and P.H. Steen, "Energy landscapes and stability to finite-amplitude disturbances for the bistable capillary bridge," Phys. Fluids, 16(8), 3063-69 (2004).

Task Progress:	2. Nitsche, M. and P.H. Steen, “Numerical simulations of inviscid capillary pinchoff,” J. Comp. Phys. 200, 299—324 (2004). 3. Hirsa, AH, CA Lopez, MA Laytin, MJ Vogel and PH Steen, “Low-dissipation capillary switches at small scales”, Appl. Phys Lett., 86, 014106, 2005. 4. Vogel, MJ, P Ehrhard and PH Steen, “The electroosmotic droplet switch: countering capillarity with electrokinetics.” PNAS, 102(34), 11974-979, 2005. 5. Bhandar, AS, PH Steen, “Liquid-bridge mediated droplet-switch: a tri-stable capillary system,” Phys. Fluids 17, 127107, 2005. Presentations. P1. Vogel, MJ, PH Steen, AS Bhandar, AH Hirsa, C.L. Cox and C. Matalanis , "Bubbles and beetles: Applications of capillary stability", 1000 Islands Fluid Mechanics Meeting, 10 May 2003, Gananoque, ON, Canada. P2. Vogel, MJ, AS Bhandar, PH Steen and AH Hirsa, "Moving liquids by design: Stability of bubble/droplet systems", Gordon Research Conference on Gravitational Effects in Physico-Chemical Systems, 30-31 July 2003, New London, CT. P3. Bhandar, AS, MJ Vogel, and PH Steen, “Stability of Capillary Systems: Moving Fluids by Design”, Gordon Conference on Gravitational Effects in Physico-Chemical Systems, 29 July 2003, Connecticut College, New London, CT. P4. Hirsa, AH, MJ Vogel, MA Laytin, JE Meyerholz, PH Steen, C. Matalanis, and A.A. Wirecki, “Capillary microswitch activation,” AIChE Annual Meeting, 16-21 November 2003, San Francisco, CA P5. Bhandar, AS, MJ Vogel and PH Steen. “Instability and switching dynamics of the bi-stable capillary bridge”, AIChE Annual Mtg, 16-21 November 2003, San Francisco, CA P6. Laytin, M. CC Lee, A Wirecki, J. Meyerholz, A. Hirsa and PH Steen, “Capillary micro-switch activation”, APS/DfD Annual Mtg, 23 November 2003, Meadowlands, NJ. P7. Vogel, MJ, M Nitsche and PH Steen, "Using curvature extrema to track the evolution of axisymmetric interfaces", APS/DfD Annual Mtg, 23 November 2003, Meadowlands, NJ. P8. Vogel, MJ, PH Steen and P Ehrhard, "Electroosmosis-activated capillary switch: Toward realization", IKET Winter Seminar, 18 February 2004, Ospizio Bernina, Switzerland. P9. PH Steen, MJ Vogel, A Bhandar and P Ehrhard, “Stability and dynamics of systems of capillary surfaces at small scales: liquid switches,” GAMM Conference, Dresden, Germany; 23 March 2004. P10. PH Steen, P Ehrhard and MJ Vogel; “Electro-kinetically triggered capillary switches.” AIChE Annual Meeting; Austin, TX, USA; November 7-12, 2004. P11. A Bhandar, PH Steen, “Stability of capillary interfaces at small scales: liquid bridge mediated droplet switches” AIChE Annual Meeting; Austin, TX, USA; November 7-12, 2004. P12. PH Steen, P Ehrhard and MJ Vogel; “Electro-kinetically triggered capillary switches,” APS/DfD Annual Meeting, Seattle, WA, USA; November 21-23, 2004. P13. M Nitsche and PH Steen; “Numerical simulations of inviscid capillary pinchoff,” APS/DfD Annual Meeting, Seattle, WA, USA; November 21-23, 2004. P14. M Ciuryla, R Chin, A Hirsa and PH Steen; “Making and breaking liquid connections with a capillary switch,” APS/DfD Annual Meeting, Seattle, WA, USA; November 21-23, 2004. P15. PH Steen, AS Bhandar, MJ Vogel and AH Hirsa, “Dynamics and stability of capillary surfaces: designing droplet switches,” Frontiers in Applied and Computational Mathematics, New Jersey Institute of Technology, Newark, NY; 13-15 May 2005. P16. PH Steen, “Countering capillarity with electro-kinetics: the electro-osmotic droplet switch.” Symposium in honor of Simon Ostrach, Cleveland, OH; October 14, 2005. P17. PH Steen, MJ Vogel and P Ehrhard; “Countering capillary pressure with electrostatic pressure at small scales,” APS/DfD Annual Meeting, Chicago, IL, USA; November 20-22, 2005.
Bibliography Type:	Description: (Last Updated: 09/04/2025)
Abstracts for Journals and Proceedings	Steen PH , Bhandar AS , Vogel MJ, Hirsa AH. "Dynamics and stability of capillary surfaces: designing droplet switches." Frontiers in Applied and Computational Mathematics, New Jersey Institute of Technology, Newark, NY, 13-15 May 2005. Frontiers in Applied and Computational Mathematics, 2005 May. , May-2005
Abstracts for Journals and Proceedings	Steen PH. "Countering capillarity with electro-kinetics: the electro-osmotic droplet switch." Symposium in honor of Simon Ostrach, Cleveland, OH; October 14, 2005. Symposium in honor of Simon Ostrach, 2005 Oct , Oct-2005
Abstracts for Journals and Proceedings	Steen PH, Vogel MJ, Ehrhard P. "Countering capillary pressure with electrostatic pressure at small scales." APS/DfD Annual Meeting, Chicago, IL, USA; November 20-22, 2005. APS/DfD Annual Meeting, 2005 Nov. , Nov-2005
Articles in Peer-reviewed Journals	Vogel MJ, Ehrhard P, Steen PH. "The electroosmotic droplet switch: countering capillarity with electrokinetics." Proc Natl Acad Sci U S A. 2005 Aug 23;102(34):11974-9. PMID: 16091462 , Aug-2005

Articles in Peer-reviewed Journals	Bhandar AS, Steen PH. "Liquid-bridge mediated droplet switch: A tristable capillary system." Phys Fluids 2005 Dec;17(12):127107. https://doi.org/10.1063/1.2149399 , Dec-2005